

Linear Algebra

Friday 2 October 2026 | Optional supplement | Student copy - no solutions

Optional S1-S4: do only the tasks your tutor selects. You do not need to finish this sheet during the main lesson.

S1 Find, correct and check

Tutor-written extension. Try for 2 minutes; then explain your correction.

A student makes the row operation shown below. Find and correct the error, solve the original system, and verify your solution in both original equations.

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & -1 & 0 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -3 & 0 \end{array} \right]$$

Correction, solution and substitution check:

S2 When does the answer change?

Tutor-written extension. Try for 4 minutes. Work over $\mathbb{R}$.

For real parameters a, b , classify all solutions of the system below. For each case, give the complete solution set when it exists.

$$\begin{aligned} x + 2y &= 3 \\ 2x + ay &= b \end{aligned}$$

If parameters have not yet been covered, use only these numerical cases:

$(a, b) = (5, 8), (4, 6), (4, 7)$.

Row reduction, cases and solution sets:

Be ready to explain why your cases cover every possibility.

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S3 is a construction task. S4 is a proof challenge: attempt it only if your tutor selects it.

S3 Build your own examples

Tutor-written extension. Try for 3 minutes. Real variables x, y .

Construct three systems, each with exactly three equations in two variables: (i) no solution; (ii) a unique solution; (iii) infinitely many solutions. Write their augmented matrices and justify each answer. Include a zero row in at least one consistent example.

Three augmented matrices and reasons:

S4 Can there be exactly two solutions?

Tutor-written proof challenge. Try for 3 minutes. Work over $\mathbb{R}$.

Let A be a real $m \times n$ matrix. Suppose $Au = b$ and $Av = b$, with $u \neq v$. For each real number t , define

$$w(t) = (1 - t)u + tv.$$

- (a) Prove that $Aw(t) = b$ for every $t \in \mathbb{R}$.
- (b) Prove that distinct values of t give distinct vectors $w(t)$.
- (c) Explain why a real linear system cannot have exactly two solutions.

Proof: show the matrix calculation and justify distinctness.
